EASTERN INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY

Practice Assignment – Quarter 3, 2026

Course Name: ISAD
Course Code: CSW 309

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LAB 02

Analysis Phase Investigation & Information Gathering

Duration: 3 periods (180 minutes)
Topic: **University Classroom Maintenance Request System**

1. Objectives

After completing this lab, students will be able to:

- Describe the current system and its main business process
- Identify problems and opportunities in the current process
- Identify key stakeholders and their roles
- Choose appropriate information gathering techniques for different information needs
- Summarize initial analysis findings clearly
- Prepare basic inputs for a later system proposal

2. Theoretical Background

In the analysis phase, systems analysts investigate the current situation, identify business needs, study problems and opportunities, identify stakeholders, gather information, and prepare analysis findings for future requirements and proposal work.

Common information gathering techniques include:

- Interviews
- Questionnaires / surveys
- Observation
- Document analysis

- Sampling existing records
- Workshops

A good analysis should be evidence-based and should connect current problems to possible system improvements.

3. Case Study (for analysis)

At a university, classroom and facility problems are currently reported and handled in a mostly informal and semi-manual way. Common issues include broken projectors, air conditioners not working, damaged lights, broken chairs, poor internet connection, faulty microphones, and other classroom equipment or facility problems that affect teaching and learning.

At present, when a problem occurs, lecturers or students usually report it by telling the faculty office, contacting the facilities staff directly, sending messages through informal chat groups, or writing notes in a paper logbook. In some cases, the same issue may be reported more than once by different people. In other cases, problems are not reported clearly because the exact room number, equipment ID, or fault description is missing.

After receiving a report, office staff or facilities staff may record the issue in different ways. Some use notebooks, some use Excel files, and some only rely on chat messages. Because there is no single standard process, the response depends on who receives the report and how urgent they think the issue is. Maintenance staff may handle urgent issues first, but there is no consistent priority rule.

As a result, some maintenance requests are delayed, repeated, or forgotten. Lecturers and students often do not know whether a problem has already been reported or when it will be fixed. Managers also find it difficult to monitor the number of requests, the most common issues, response time, and staff workload.

The university is considering a web-based maintenance request system that allows users to report issues, track request status, assign tasks to maintenance staff, and generate summary reports for management.

4. Tasks

Study the current classroom maintenance request process, identify business problems and stakeholders, choose suitable information gathering techniques, and prepare initial proposal inputs for a future system.

4.1 Study the Current System

Complete the table below:

Item	Student Answer
Current system type	Informal, semi-manual, fragmented system
Main process	Report issue → Someone receives (office staff, facilities, chat group, logbook) → Record in own way → Handle if urgent → No consistent follow-up
Main records / communication methods used	Paper logbook, Excel files, chat messages (Zalo/MS Teams), verbal reports, notebooks
Main people / units involved	Lecturers, students, faculty office staff, facilities/maintenance staff, department managers
Common difficulties in the current process	Duplicate reports, missing room/equipment details, no tracking, forgotten requests, inconsistent priority, no feedback to reporter

4.2 Identify Problems and Opportunities

Define:

- 5 business problems
- 3 improvement opportunities

Type	Description
Problem 1	No standard reporting process → issues are forgotten or delayed
Problem 2	Duplicate reports due to lack of centralized tracking
Problem 3	Missing critical information (room number, equipment ID, fault description)
Problem 4	No way for reporters to check request status
Problem 5	Managers cannot monitor response time, workload, or common issues
Opportunity 1	Implement a web-based system to centralize all maintenance requests
Opportunity 2	Use automated priority rules and assignment to improve response time
Opportunity 3	Generate reports for management to identify trends and allocate resources effectively

4.3 Identify Stakeholders

Identify main stakeholders and their roles (at least 4 stakeholders)

Stakeholder	Role in Current Process	Main Concern / Interest
Lecturers	Report classroom issues informally	Quick resolution, minimal disruption to teaching
Students	Report issues (e.g., broken chair,..)	Comforable learning environment
Facilities Staff	Receive, priority, and fix issues manually	Clear task assignment, reduce repeated work
Faculty office staff	Relay message, sometimes record issues	Not being overloaded, knowing request status
University management	Monitor maintenance performance	Data visibility, cost control, service quality

4.4 Choose Information Gathering Techniques

Information Needed	Best Technique	Why?
Lecturer and student frustrations	Survey + Questionaire	Can reach many users quickly; anonymous feedback
Actual reporting and handling steps	Interview + Observation	Observe real workflow and ask follow-up questions to clarify gaps
Required maintenance details and request fields	Document analysis + Sampling	Review existing logbooks, Excel files, chat logs to find missing fields
Common repeated issues and delays	Sampling existing records + Interview	Count frequently from records, ask staff why delays happen
Agreement on future improvement priorities	Workshop	Bring key stakeholder together to resolve conflicts and align expectations

4.5 Problem Description and System Request

- Summarize the problem (why the system is needed).

The current maintenance request process is informal, fragmented, and lacks standardization. Request are lost, duplicated, or delayed because there is no central tracking, consistent priority rules, or feedback mechanism. Managers have no visibility into maintenance performance.

- Write the goal and 2–3 expected benefits

To design a web-based classroom maintenance request system that enables easy reporting, status tracking, task assignment, and management reporting.

Expected benefits:

- Reduce forgotten or delay maintenance requests

- Improve transparency for lecturers or students

- Enable data-driven decisions for management

4.6 Functional Requirements

List 5 main functions the system must support

1. Users can submit a maintenance request with room number, equipment type, fault description, and photo attachment.

2. System automatically assigns a priority level based on rules

3. Facilities staff can update request status online

4. Reporters can track their request status online

5. Management can generate reports

4.7 Prepare Information Gathering Instruments

Choose ONE of the following:

- Write 5 interview questions for facilities staff

  - Can you walk me through how to receive and handle a maintenance request?

  - What information do you often find missing when someone reports a problem?

  - How do you decide which request to handle first?

  - Have you worked on a request that was already fixed by someone else?

  - What would make your job easier in new system?

- Write 5 survey questions for lecturers/students

5. Assessment (100 points total)

Criterion	Points	What to Look For
Understanding of current system	15	identifies the semi-manual / fragmented process correctly
Problems and opportunities	20	problems are relevant, opportunities are realistic
Stakeholder analysis	10	identifies suitable stakeholders and their concerns
Technique selection	15	technique matches information need with valid reason

Problem description, goal, and benefits	15	explains need clearly; goal and benefits are logical
Functional requirements	15	lists 5 suitable high-level system functions
Information gathering instrument	10	questions/items are clear, relevant, and usable

Home completion:

1. Initial Analysis Findings

Write a short paragraph of 150–200 words summarizing:

- The most serious problem
- The most important stakeholder
- The most useful information gathering technique
- The main areas for improvement

The primary issue in the current classroom maintenance approach is the absence of a unified and standardized system. Reports are made via chat, logbooks, or in person, which results in duplicated, overlooked, or incomplete requests. The key stakeholders are the facilities staff, whose daily tasks are negatively impacted by ambiguous requests and a lack of prioritization. A valuable method for gathering information is to conduct interviews with facilities personnel alongside surveys of teachers and students, as this reflects both practical situations and user discontent. Essential improvement areas include: establishing a web-based request platform with mandatory fields (room, equipment, description), introducing automatic priority settings and status updates, and enabling management reports to keep track of response times and frequent problems. Focusing on these aspects will minimize delays, enhance transparency, and assist the university in better distributing maintenance resources.

2. Proposal Input Draft

Based on the findings, write:

- 1 problem statement
- 3 project objectives
- 3 items in scope, 2 items out of scope

Problem statement:
The current informal and semi-manual maintenance request process leads to lost, duplicated, and delayed repairs, with no visibility for reporters or management.

Project objectives:

- Implement a web-based system to centralize all classroom maintenance requests
- Reduce average request response time by 50% within 3 months of launch
- Provide management with monthly reports on request volume, response time, and common issues

In scope:

- Online request submission with required fields and photo upload
- Status tracking for reporters
- Basic priority rules and task assignment for facilities staff

Out of scope:

- Integration with existing university ERP or room booking system
- Mobile app development (web-based only)